

The Pedestal Correction

Why the Pigment Index reads about 29 % too high, where the error comes from, and what a single subtraction would do about it.

Why this document exists. On 2026-08-01 three preparations of the same green oil returned three different verdict numbers — 14.28, 12.74 and 13.04 — a spread of 10.8 % on an oil that had not changed. Chasing that down produced a measurement of a quantity the specification had only ever derived on paper, and with it a correction that would remove most of the spread. This document explains the whole chain from first principles, so that the proposal can be judged rather than believed.

Its companions. *Capture Fidelity* covers the instrument — how a webcam becomes a spectrum. *Light, Pigment and Solvent* covers the sample. *From Spectrum to Verdict* covers the arithmetic that turns an absorbance curve into a verdict. **This document is a correction to that arithmetic**, and assumes only chapter 5 of it.

How to read it. Chapters 2–4 build the problem from nothing and contain no algebra beyond a subtraction. Chapter 5 is the one piece of real algebra and is worth the effort — everything after it is consequence. Chapter 8 says what the corrected number physically IS; chapter 9 is the first real test of it. Chapter 10 lists every assumption in one place. **Chapter 12 is the chapter that says why this may be wrong**, and should be read before anybody acts on it.

Status. The measurement is real. The correction is a **proposal** — fitted on one oil, confirmed out-of-sample on a second, and shown by the same test NOT to survive a rig rebuild.

CONTENTS

1 · The claim, in one page

2 · What the instrument actually hands us

3 · The pedestal, and why a baseline is subtracted at all

4 · What a straight line cannot remove — and which way it errs

5 · Where the error lands — and why the denominator suffers

6 · Measuring r_Q — a test with no free parameters

7 · The correction, worked end to end

Worked example — Kiendler set A, the over-dilute one

The same, for every set on record

Why one cannot simply concentrate the problem away

8 · What the corrected number IS, physically

8.1 Three layers of grounding

8.2 ★ Why invariance and sensitivity arrive together

8.3 △ Three places the physics is NOT settled

8.4 What may therefore be claimed

9 · ★ The out-of-sample test — the first real evidence, and the limit it finds

What the good row is worth

What the two bad rows are worth

The reading that fits all three rows

10 · Every assumption, in one place

△ What the two-oil agreement is worth — considerably less than it looks

What the evidence actually is, in descending order of weight

⇒ The verdict splits

11 · What the correction buys, and what it costs

12 · △ The chapter that argues against the proposal

The ranking test, and how it fails

Three other ways this could be wrong

13 · What would settle it

Appendix A — every number

Appendix B — reproduction

Appendix C — where this sits in the specifications

1 · The claim, in one page

The instrument produces an absorbance curve. The verdict is a **ratio of two heights** taken off that curve, after a straight line has been subtracted from it:

$$M = \frac{B_{\text{Soret}}}{B_Q}$$

B_{Soret} is the pigment's blue absorption band (440–460 nm); B_Q is its much weaker green-yellow band (560–580 nm). Green oil scores high, brown oil low, and the shipped threshold is 10.6.

The problem. The straight line that gets subtracted is an approximation, and it does not remove exactly the right amount. What it leaves behind at the Q band we call r_Q . Because B_Q is a small number, a small leftover is a large fraction of it — and B_Q is the **denominator**, so the whole ratio inherits that fraction.

The measurement. $r_Q = -0.0246 \pm 0.0037 \text{ A}$, obtained from ten runs of one oil. A second oil gives a consistent $-0.0212 \pm 0.0193 \text{ A}$, but that interval contains zero and adds very little — chapter 10 weighs it honestly, and the real confirmation is chapter 9's out-of-sample test. r_Q is **negative**, meaning the correction subtracts slightly too much.

The size. At the standard recipe this inflates the verdict number by about **29 %**; on an over-dilute sample it reached **43 %**.

The proposal. Put the leftover back before dividing:

$$M_{\text{corrected}} = \frac{B_{\text{Soret}}}{B_Q - r_Q}$$

One subtraction, using quantities every run already computes. Applied to the three preparations that started this, their spread falls from **10.8 % to 2.5 %**.

What it costs. The numbers move to a different scale — green lands near 10.0 instead of 12–14, brown near 7.6 instead of 9.3 — so **the threshold 10.6 becomes meaningless and must be re-derived**.

What it assumes. That r_Q is a *constant*. That is the whole risk. Chapter 9 tests it out-of-sample for the first time, and chapter 12 shows a concrete case where it changes the answer.

2 · What the instrument actually hands us

Everything below starts from one curve: **absorbance against wavelength**, written $A(\lambda)$.

Absorbance is defined from two captures — the reference (pure solvent, no oil) and the sample:

$$A(\lambda) = \log_{10} \frac{R(\lambda)}{S(\lambda)}$$

$A = 0$ means the sample passes as much light as the blank. $A = 1$ means it passes a tenth of it. $A = 2$, a hundredth. The scale is logarithmic, so **high absorbance means very little light is getting through** — and there is a ceiling, because once the transmitted signal falls into the sensor noise the number stops meaning anything.

From that curve the plugin reads **four wavelength windows**:

window	range	what lives there
Soret band	440–460 nm	the pigment's strong blue absorption
near anchor	520–540 nm	chosen to be <i>quiet</i> — little pigment
Q band	560–580 nm	the pigment's weak second absorption
far anchor	600–630 nm	also chosen as quiet

The two **anchors** are not measurements of the pigment. They exist to answer a different question, which is the subject of the next chapter.

A note on band values. Each window's value is the mean absorbance over the points inside it. Where this document quotes a set's value it is the **mean over that set's runs**. One consequence worth knowing when checking the arithmetic: the mean of a ratio is not exactly the ratio of the means, so recomputing M from the quoted B_{Soret} and B_{Q} reproduces the quoted M to about 0.05 %, not exactly.

3 · The pedestal, and why a baseline is subtracted at all

A cuvette of diluted oil does not only *absorb* light. It also **scatters** it — off undissolved oil droplets suspended in the alcohol. Scattered light misses the camera just as absorbed light does, so the instrument cannot tell the two apart. Both arrive as "absorbance".

The result is that the measured curve sits on a **pedestal**: a broad, smooth background that is present at every wavelength and has nothing to do with the pigment.

$$A_{\text{measured}}(\lambda) = A_{\text{pigment}}(\lambda) + P(\lambda)$$

The pedestal P is a nuisance and it is large. Earlier work measured it at roughly **7 % of the Soret band but 52–61 % of the Q band** — because the Q band is intrinsically weak, the same background is a far bigger share of it.

The standard remedy is to estimate P from places where the pigment is quiet, and subtract it. That is what the two anchor windows are for: fit a straight line through 520–540 nm and 600–630 nm, and subtract that line from the whole curve. Everything called $B_{\dots}$ in this document means a band value **after** that subtraction.

This works well. It is the reason the shipped metric survives jar-tilt events that send the uncorrected ratio wandering by 29 %, and nothing here argues against doing it.

But a straight line is a straight line, and the pedestal is not straight.

4 · What a straight line cannot remove — and which way it errs

Scattering does not fall off linearly with wavelength. It falls off roughly as a power law, $P \propto \lambda^{-n}$, which is a **convex** curve: steep in the blue, flattening toward the red.

Now the geometry that decides everything:

- We fit the line through two windows, at **520–540 nm** and **600–630 nm**.
- A straight line through two points on a convex curve is a **chord**.
- **A chord lies above the curve everywhere between its endpoints.**
- The Q band, **560–580 nm**, lies exactly between them.

So at the Q band the fitted line sits **above** the true pedestal, and subtracting it removes the pedestal **plus a slice of genuine pigment signal**.

Write the leftover as r , defined as what the line fails to remove — negative where the line over-subtracts:

$$B_X = A_{\text{pigment},X} + r_X$$

This predicts the sign before we measure anything: r_Q must be negative. It is a real prediction and it could have come out wrong. Measured, $r_Q = -0.0246$ A.

Why the Soret band escapes. 440–460 nm lies *outside* the two anchors, not between them. There the chord runs below the curve, and in any case the Soret band is ten to twenty times taller, so a leftover of 0.025 A is a rounding error against 1.1 A. Throughout this document r_{Soret} is treated as negligible; chapter 10 lists that as an assumption.

5 · Where the error lands — and why the denominator suffers

This is the one piece of algebra in the document. Take it slowly; everything afterwards follows.

Write the true pigment absorbance in each band as concentration c times path length ℓ times that band's extinction coefficient e :

$$A_{\text{pigment,Soret}} = c \cdot \ell \cdot e_{\text{Soret}} \quad A_{\text{pigment,Q}} = c \cdot \ell \cdot e_{\text{Q}}$$

Substituting into chapter 4's definition, what we actually measure is:

$$B_{\text{Soret}} = c \cdot \ell \cdot e_{\text{Soret}} + r_{\text{Soret}} \quad B_{\text{Q}} = c \cdot \ell \cdot e_{\text{Q}} + r_{\text{Q}}$$

The **true** index — the one that depends only on which pigment is present, not how much — is the ratio of the extinction coefficients:

$$M_{\infty} = \frac{e_{\text{Soret}}}{e_{\text{Q}}}$$

Now form the index we actually compute, dropping r_{Soret} as negligible:

$$M = \frac{B_{\text{Soret}}}{B_{\text{Q}}} = \frac{c \cdot \ell \cdot e_{\text{Soret}}}{c \cdot \ell \cdot e_{\text{Q}} + r_{\text{Q}}}$$

Divide top and bottom by $c \cdot \ell \cdot e_{\text{Q}}$:

$$M = M_{\infty} \cdot \frac{1}{1 + \frac{r_{\text{Q}}}{c \cdot \ell \cdot e_{\text{Q}}}}$$

Read that denominator. The error term is not r_{Q} on its own — it is r_{Q} **divided by the pigment signal**. Three consequences follow immediately, and they are the whole story:

1. **The error is a *fraction*, not an offset.** A fixed leftover does more damage to a weak sample than a strong one.
2. **It grows without limit as the sample is diluted.** Halve the concentration and you double the error. This is why over-dilute preparations misread — see figure 2.
3. **r_{Q} sits in the denominator, so its sign is inverted in the result.** r_{Q} negative makes the whole expression **larger**. The index reads **high**, not low.

Why the denominator and not the numerator. The same absolute error e shifts a ratio by e/B in each band, so the damage ratio is simply how much bigger one band is than the other:

$$\frac{B_{\text{Soret}}}{B_Q} = \frac{1.1160}{0.0856} \approx 13$$

An error in **B_Q** is about thirteen times more damaging than the same error in **B_{Soret}**. That is the entire reason this document is about the Q band.

Finally, rearranging the same expression into the form used from here on:

$$M = M_{\infty} \cdot \left(1 - \frac{r_Q}{B_Q} \right)$$

The bracket is the **inflation factor**. With **r_Q** negative it is greater than 1.

6 · Measuring **r_Q** — a test with no free parameters

We now need **r_Q** from data. The trick is to avoid needing to know the concentration at all — which matters, because the session that produced this data proved the concentration was not reliably known.

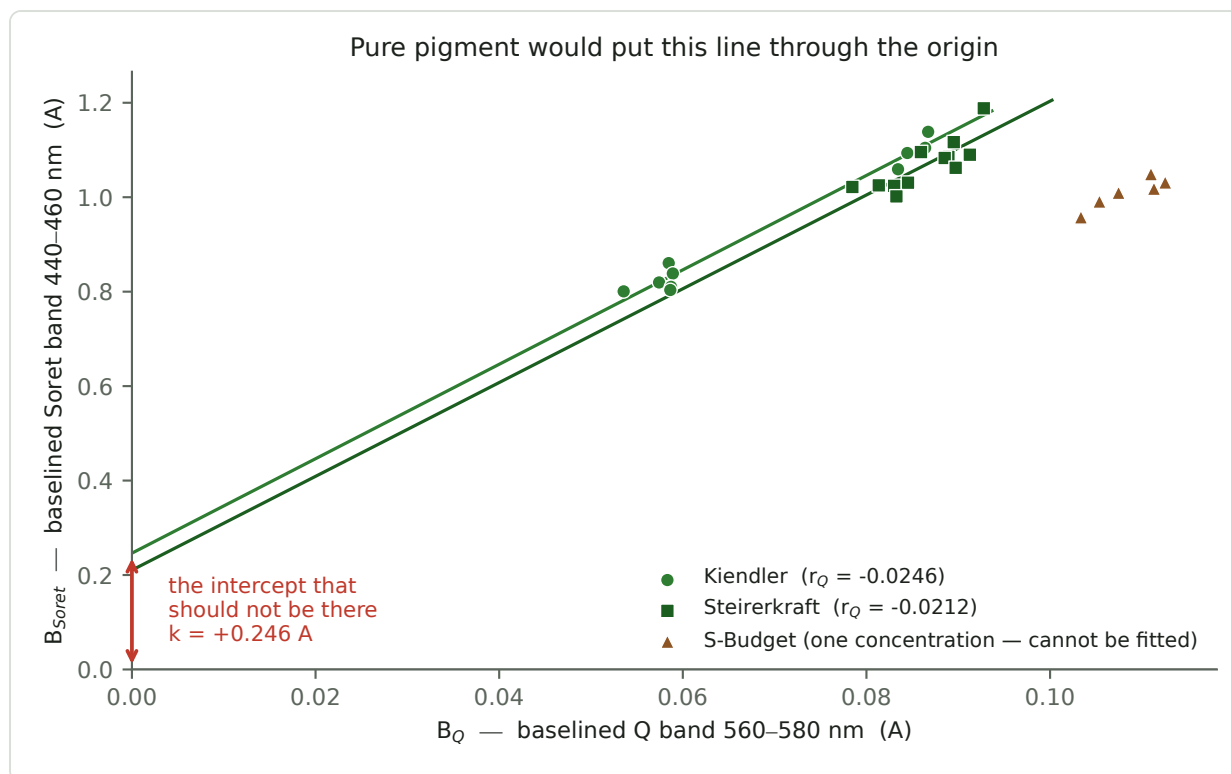
Return to chapter 5's two measured quantities and eliminate **c**:

$$B_{\text{Soret}} = M_{\infty} \cdot B_Q + (r_{\text{Soret}} - M_{\infty} \cdot r_Q)$$

That is the equation of a **straight line** relating two things we measure directly. Vary the concentration however sloppily you like: every run must fall on it.

And here is the test. If there were no leftover at all — a perfect baseline — then both **r** terms are zero, the bracket vanishes, and **the line passes through the origin**. That makes intuitive sense: with no pedestal left, when one band's pigment signal is zero so is the other's.

So: plot **B_{Soret}** against **B_Q**, one point per run, and look at where the line crosses. The intercept is the leftover. **No fitted concentration, no assumed recipe, no free parameters.**



The straight-line test

Fitted at run level, so the intercept carries an honest standard error:

oil	runs	slope = M_∞	intercept k	t	$\Rightarrow r_Q = -k / M_\infty$
Kiendler	10	9.998 ± 0.502	$+0.2463 \pm 0.0351$	7.01	$-0.0246 \pm 0.0037 \text{ A}$
Steirerkraft	12	9.930 ± 2.149	$+0.2103 \pm 0.1860$	1.13	$-0.0212 \pm 0.0193 \text{ A}$
S-Budget	6	—	—	—	one concentration only — cannot be fitted

Reading the table.

- The intercept is **not zero**, at 7 standard errors for Kiendler. The baseline leaves something behind, and the amount is what chapter 4 predicted in sign.
- The **slope** is the pedestal-free index M_∞ , and both green oils give ≈ 9.9 – 10.0 .
- Steirerkraft's fit proves nothing on its own** ($t = 1.13$). Its B_Q values span only 0.0785–0.0928 — **17 % of the mean against Kiendler's 48 %**, too short a lever to locate an intercept — and its 95 % interval on r_Q , $-0.0590 \dots +0.0166$, **contains zero**. It is quoted because its point estimate independently agrees, not because it is significant. Chapter 10's A1 weighs how little that is worth.
- The brown oil cannot be used at all**. It exists at one concentration, and you cannot fit a line through points that share an x-value.

Where Kiendler's leverage came from — and the irony in it. Its B_Q spans 0.0536–0.0867, wide enough to fit, *only because one of its three preparations was accidentally too dilute*. The botched sample is what made the measurement possible.

7 · The correction, worked end to end

Start from chapter 5's result and solve for the quantity we actually want:

$$M_{\infty} = \frac{M}{1 - \frac{r_Q}{B_Q}} = \frac{B_{\text{Soret}}}{B_Q - r_Q}$$

It collapses to a single subtraction. $B_Q - r_Q$ is nothing other than the pigment part of the Q band with the baseline's over-subtraction put back. Then divide as before.

Worked example — Kiendler set A, the over-dilute one

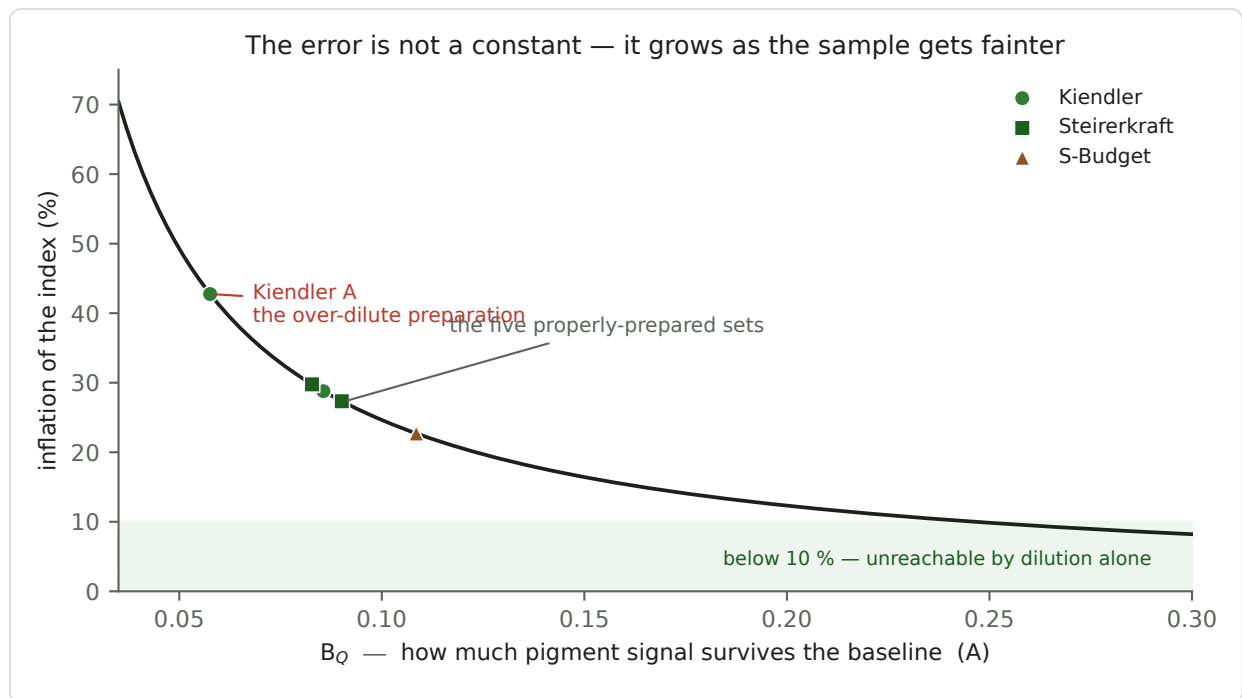
step		
1	read the baselined Soret band	$B_{\text{Soret}} = 0.8221 \text{ A}$
2	read the baselined Q band	$B_Q = 0.0576 \text{ A}$
3	the shipped index	$0.8221 / 0.0576 = \mathbf{14.27}$
4	put the leftover back	$B_Q - r_Q = 0.0576 - (-0.0246) = \mathbf{0.0822 \text{ A}}$
5	divide again	$0.8221 / 0.0822 = \mathbf{10.00}$
6	how inflated step 3 was	$14.27 / 10.00 = \mathbf{1.43}$, i.e. +43 %

Step 4 is the whole correction. Everything else was already being computed.

The same, for every set on record

set	B_Q	inflation	M shipped	M corrected
Kiendler A (<i>over-dilute</i>)	0.0576	+42.8 %	14.279	10.00
Kiendler B	0.0849	+29.0 %	12.740	9.88
Kiendler C	0.0856	+28.8 %	13.039	10.13
Steirerkraft B	0.0828	+29.8 %	12.489	9.62
Steirerkraft C	0.0901	+27.3 %	12.251	9.62
S-Budget D (<i>brown</i>)	0.1085	+22.7 %	9.303	7.58

The three Kiendler preparations spread 10.8 % before and 2.5 % after. That is the result the proposal rests on.



Inflation against pigment signal

Why one cannot simply concentrate the problem away

The inflation is $|r_Q| / B_Q$, so working at higher concentration shrinks it. How much higher?

target inflation	needs $B_Q \geq$	relative to today's 0.0856
30 %	0.082	1.0× — <i>this is where we are</i>
20 %	0.123	1.4×
10 %	0.246	2.9×
5 %	0.493	5.8×

There is no room. At today's recipe the 440–447 nm bins already read 2.0–2.6 DN against a reference near 88 — they are dark, and previous work established they are not measurements. Working 1.4× stronger pushes more of the Soret band into that floor; 2.9× is out of the question.

⇒ **The concentration route is closed.** Either correct the number arithmetically, or reduce r_Q itself by making the sample physically clearer — filtration, or a solvent that dissolves rather than suspends.

8 · What the corrected number IS, physically

Chapter 7 produced a quantity that is more stable than the one we had. **Stability is not meaning.** A number can be beautifully reproducible and still measure nothing in particular, so this chapter asks what M_∞ corresponds to in the sample — and, just as importantly, where that correspondence stops being established.

8.1 Three layers of grounding

Beer-Lambert — the derivation itself. Every step of chapter 5 is $A = \epsilon \cdot c \cdot \ell$ plus algebra.

$M_\infty = e_{\text{Soret}}/e_{\text{Q}}$ is a ratio of **extinction coefficients**: quantities belonging to the molecule, not to the sample, the operator or the instrument.

Porphyrin spectroscopy — why those two bands exist at all. The Soret and Q bands are not windows we invented; they are the standard electronic structure of a porphyrin macrocycle. In the four-orbital picture the Soret transition is intensely allowed and the Q transitions are weak — which is precisely *why* the denominator of this metric is small and fragile, and therefore why this whole document is about the Q band rather than the Soret one.

It also supplies the chemistry the verdict depends on. The pigment is **protochlorophyll**, a porphyrin. Losing its central magnesium — which is what ageing and roasting do, giving **protopheophytin** — lowers the macrocycle's symmetry from D_{4h} to D_{2h}, splits the Q bands and redistributes intensity among them. **A Soret-to-Q ratio therefore probes metallation state**, which is the chemical difference between a fresh oil and a degraded one.

Scattering — and a prediction that could have failed. Chapter 4 did not *fit* the sign of r_{Q} ; it **predicted** it, from λ^{-n} scattering being convex and the Q band lying between the two anchors. Measured, $r_{\text{Q}} = -0.0246$. **Had it come out positive the model would have been dead.** A prediction that risks refutation and survives is worth substantially more than a good fit, and it is the main reason to believe the mechanism is the one described rather than a curve that happens to match.

8.2 ★ Why invariance and sensitivity arrive together

The oil is not one pigment. It is a **mixture** of intact protochlorophyll and its degradation products, and the entire point of the verdict is that their proportions differ between a fresh oil and an old one. For a mixture the corrected index becomes

$$M_\infty = \frac{\sum_i c_i \cdot e_{\text{Soret},i}}{\sum_i c_i \cdot e_{\text{Q},i}}$$

Now apply the two operations that matter, one after the other:

Dilute the sample — multiply every c_i by the same factor k :

$$\frac{\sum_i k \cdot c_i \cdot e_{\text{Soret},i}}{\sum_i k \cdot c_i \cdot e_{\text{Q},i}} = \frac{\sum_i c_i \cdot e_{\text{Soret},i}}{\sum_i c_i \cdot e_{\text{Q},i}}$$

k cancels. **Dilution-invariant — whatever the composition is.**

Degrade the sample — change the *proportions* between species. The sums no longer scale together, because the species have different ϵ at the two bands, so the ratio moves. **Speciation-sensitive.**

Both properties fall out of the same three lines. Neither was tuned in, and they are exactly the pair the application needs: blind to how much oil went into the alcohol, sensitive to what the pigment has become. It is also the measured result — `SPEC_capture_quality.md` §16.13.9 refuted single-species scaling at $d = 10.26$, so the mixture reading is not an assumption but an observation.

8.3 $\triangle$ Three places the physics is NOT settled

(a) The 560–580 nm assignment is open — and it is the denominator of the verdict. Two candidates, which are not equivalent:

candidate	what the metric would then be
Q(1,0) — vibronic satellite of the intact, Mg-bearing pigment	a comparison of two bands of the <i>same</i> molecule
Qx of protopheophytin — the metal-free degradation product	a literal intact ÷ degraded ratio, with a far stronger justification

Evidence gathered 2026-07-31 leans to the second: `A_Q` is *equal* across the classes while `A_Soret` differs by 9 %, and the 572 nm feature is *stronger in the brown oil* — a band that grows as the oil degrades is hard to attribute to the intact molecule. **Not proven.** No source we hold assigns this band, and the comparison is between two different bottles rather than one oil before and after demetallation. The **acid test** — split one oil, demetallate half, re-measure — settles it in an afternoon (`SPEC_capture_quality.md` §16.13.8).

(b) Our windows are not the bands, so the number is not literature-comparable. 440–460 nm is the right-hand *slope* of the Soret band; its peak may lie below the 440 nm edge of the ROI entirely. So $M_\infty \approx 10.0$ is **our windows' number**, not a molar ratio anyone could look up or reproduce on another instrument. **It is dilution-invariant but not absolutely calibrated** — a relative index. Putting it on the published scale would mean measuring in 80 % acetone, the literature standard for protochlorophyll; recorded, not scheduled.

(c) Beer-Lambert is strained exactly where the numerator lives. At the Soret band the sample transmits ~6.5 % and the blue edge is already dark, so linearity is weakest at the very band the numerator is built from. This is assumption A4 in the next chapter, and it is not a small print item.

8.4 What may therefore be claimed

Defensible today: that the *invariance* is a consequence of Beer-Lambert and not of tuning; that the *sensitivity to degradation* reflects real macrocycle chemistry; and that the sign of the pedestal residual was predicted from scattering physics before it was measured.

Not defensible today: that M_∞ is a molecular constant, that it can be compared with a published figure, or that the metric is a proven intact-to-degraded ratio. **The first two are matters of calibration convention; the third is one afternoon's experiment away.**

9 · ★ The out-of-sample test — the first real evidence, and the limit it finds

Everything so far was fitted. This chapter tests the result against data that had no part in producing it.

The setup. `r_Q` was fitted on **one oil, Kiendler, on the post-rebuild rig**. The archive holds several *within-oil dilution pairs* — the same oil measured at two deliberate strengths. Each is an independent test with an unambiguous prediction:

If the correction is right, it must push each pair's dilution slope **toward zero**. If it pushes any of them away, something is wrong with it.

No fitting is involved. `r_Q` is simply carried over and applied.

pair	rig state	span	shipped s	corrected s	
green oilK → oilL	pre-rebuild	1.50×	+0.01	+0.28	✗ worse
brown oilN → oilM	pre-rebuild	1.50×	+0.12	+0.26	✗ worse
green 0729B → 0729C	post-rebuild	1.17×	-0.12	-0.00	✓ fixed

What the good row is worth

On the post-rebuild pair the correction takes the dilution slope from -0.12 to -0.00. That is the proposal doing exactly what it claims, on data it never saw — and it is **not circular**: `r_Q` was measured on *Kiendler*, this pair is *Steirerkraft*. It is the first evidence that the residual is a property of the instrument rather than of one oil, and it is the strongest single result in this document.

What the two bad rows are worth

On both pre-rebuild pairs the correction makes invariance markedly worse — and those pairs are the archive's principal evidence that the shipped metric is dilution-invariant at all. A correction that destroys that is not a small matter.

The reading that fits all three rows

`r_Q` transfers between oils, but not across a rig rebuild. That is physically sensible: the residual is the curvature of the scattering pedestal as the optics present it, and the 2026-07-29 rebuild changed the optics. Three consequences follow, and they are now part of the proposal:

1. **`r_Q` is a per-rig-state calibration constant**, not a constant of nature. It must be **re-measured after any mechanical change** to the instrument.
2. **It must never be applied retroactively.** Correcting archived pre-rebuild results with today's `r_Q` produces worse numbers than leaving them alone.
3. **Assumption A1 splits in two.** "Same for every sample" is now supported *within one rig state* and refuted *across rig states*. The next chapter reflects that.

⚠ **This does not rescue the proposal, and it is not meant to.** One good row is one good row. But it is the first test the correction could have failed outright and did not, and it converts A1 from an untested hope into a bounded, testable claim.

10 · Every assumption, in one place

The correction is one subtraction, but it rests on six statements. They are listed in descending order of how much damage they would do if false.

A1 — r_Q is the same for every sample within one rig state. *This is the load-bearing one, and chapter 9 has now split it in two.*

r_Q describes the curvature of a particular sample's pedestal, and pedestals come from suspended droplets, which differ between preparations. If r_Q varied sample to sample, subtracting one fixed number would not be a correction but a new error.

△ What the two-oil agreement is worth — considerably less than it looks

The two fitted values are -0.0246 and -0.0212 , 16 % apart, and it is tempting to read that as confirmation. It is not:

	Kiendler	Steirerkraft
B_Q span, as % of its mean	48 %	17 %
r_Q	-0.0246 ± 0.0037	-0.0212 ± 0.0193
95 % interval	$-0.0319 \dots -0.0173$	$-0.0590 \dots +0.0166$

Steirerkraft's interval contains zero. That oil on its own cannot establish that a residual exists at all, let alone that it matches. And the formal comparison — difference -0.0035 ± 0.0196 , $p = 0.86$ — is close to vacuous: **the smallest difference it could detect is 0.0385, which is 156 % of r_Q itself. It would miss a doubling.**

The cause is the **lever arm**. r_Q is an extrapolation back to $B_Q = 0$, and Steirerkraft's B_Q span is barely a third of Kiendler's. Same rig, same protocol, same operator — only the concentration range differs, and that alone makes the interval five times wider.

△ **A tempting shortcut that does not work, recorded so nobody re-derives it.** One can instead ask "what r_Q would make Steirerkraft's two fills read the same?" That gives **-0.0242 — a striking 2 % from Kiendler's value.** Bootstrapped over its runs, its 95 % interval is **$-0.114 \dots +0.010$ — wider than the regression it was meant to improve on.** The apparent precision was entirely an artefact of quoting a point estimate without an error bar.

What the evidence actually is, in descending order of weight

1. **The consequence test (chapter 9).** Carrying Kiendler's r_Q onto Steirerkraft drives its dilution slope from -0.12 to **-0.00** , and its two fills from a -1.85 % gap to **$+0.02$ %**. Testing what the constant *does* has far more power than comparing two blurry parameters, because the fills either land on top of one another or they do not.
2. **Reproducibility within one oil.** Kiendler's two independent preparation contrasts give **-0.0206** ($-0.0287 \dots -0.0138$) and **-0.0288** ($-0.0379 \dots -0.0213$) — overlapping intervals, bracketing the pooled -0.0246 . So r_Q reproduces across independent preparations of one oil.
3. **The two-oil point-estimate agreement.** Real, and nearly weightless, for the reasons above.

⇒ The verdict splits

- **Across OILS, within one rig state: supported** — on items 1 and 2, not on item 3.

- **Across RIG STATES: refuted.** The same constant applied to pre-rebuild pairs makes invariance *worse*, $+0.01 \rightarrow +0.28$ and $+0.12 \rightarrow +0.26$.

⇒ **r_Q is a per-rig-state calibration constant.** It must be re-measured after any mechanical change and never applied retroactively. Chapter 12 shows the assumption still changing an answer *within* a rig state, so "supported" is not "settled".

A2 — r_{Soret} is negligible. The leftover at the Soret band is treated as zero. Justification: the Soret band lies outside the anchors and is 13× taller, so the same leftover is 13× less important. **Not separately measured** — the fit cannot distinguish r_{Soret} from $M_{\infty} \cdot r_Q$, because only the combination appears in the intercept. If r_{Soret} is not zero, the quoted r_Q absorbs it and is biased.

A3 — the pedestal is convex. Chapter 4's sign prediction depends on it. Scattering theory says convex; the measured sign agrees. If some sample's pedestal were concave, the correction would push the wrong way.

A4 — Beer-Lambert holds in both bands. That absorbance is proportional to concentration, with no saturation. At the Soret band the transmitted signal is already at ~6.5 % and the blue edge is dark, so this is **weakest exactly where the numerator lives**. Stray light in a saturated band compresses absorbance, which would appear as a concentration-dependent error of its own.

A5 — one pigment species dominates each band. The model has a single e_{Soret} and a single e_Q. The oil actually contains protochlorophyll *and* its degradation product, and the whole point of the verdict is that their proportion changes. The index is therefore already a mixture ratio; the correction does not change that, but M_{∞} should be read as "the pedestal-free index", not as a physical extinction ratio.

A6 — the two anchor windows are quiet. They are not, entirely: the far anchor at 600–630 nm sits on the pigment's Q_y flank. Pigment contamination of an anchor scales with concentration, so it is **harmless to the argument here** — it cannot produce a concentration-independent intercept. What the intercept measures is specifically the part that does *not* scale.

11 · What the correction buys, and what it costs

Buys — the spread collapses. Three preparations of one oil: 10.8 % → 2.5 %.

Buys — the number stops depending on preparation quality. Today an unusually clear or unusually cloudy sample shifts the verdict by ~10 % with nothing to warn the operator. That is the property that makes a measurement **transferable between operators and sites** — which matters more for a laboratory partner than precision does.

Costs — the threshold must be re-derived. Everything moves down: greens from 12–14 to ≈ 9.6–10.1, brown from 9.3 to ≈ 7.6. **The shipped $T = 10.6$ would sit above both green oils and reject them.** A corrected threshold would land near 8.6, but it must be *derived*, not scaled.

Costs — one more constant to maintain. r_Q becomes a calibration parameter with a provenance, an uncertainty and a review schedule.

Does not buy — better discrimination. Between green and brown the correction changes very little, because both classes are inflated by similar amounts. **It is an accuracy and transferability fix, not a sensitivity fix.** Anyone hoping it will sharpen the green/brown call should not expect it to.

A caution about T that applies whether or not the correction is adopted. The present threshold was calibrated on inflated numbers. If the sample is ever made physically cleaner — a filter, a better solvent — r_Q shrinks toward zero, every reading falls by up to 29 %, and **$T = 10.6$ stops being valid with no error message and no visible symptom.** That risk exists today.

12 · ▲ The chapter that argues against the proposal

Read this before acting on the document.

The ranking test, and how it fails

The correction appears to confirm something independent: the operator's own visual judgement that the Kiendler oil is slightly greener than the Steirerkraft one. Corrected, Kiendler reads 10.00 and Steirerkraft 9.62 — Kiendler greener by 3.9 %, agreeing with the eye.

That agreement is an artefact of assumption A1. The 9.62 was obtained by applying *Kiendler's* r_Q to Steirerkraft. Using each oil's **own** measured r_Q instead:

	corrected with shared r_Q	corrected with each oil's own r_Q
Kiendler	10.00	10.00
Steirerkraft	9.62	9.93
S-Budget	7.58	7.58
Kiendler vs Steirerkraft	+3.9 %	+0.7 %

Under the second reading the two green oils are indistinguishable, and the apparent confirmation of the visual ranking disappears entirely. The data cannot currently decide between the two readings, because Steirerkraft's own r_Q is known only to ± 0.019 — an error bar wide enough to contain both.

⇒ **The correction must not be adopted because it agrees with the eye. On this evidence, whether it agrees with the eye is a free parameter.**

Three other ways this could be wrong

- **The fit is dominated by one contrast.** Kiendler's intercept is driven by set A sitting apart from sets B and C. That is one comparison with $n = 6$ against $n = 4$, not ten independent points.
- **The two axes share a spectrum.** B_{Soret} and B_Q come from the same measurement, so noise common to both inflates the *slope*. The **intercept** is the claim and is more robust — but the concern is real and unquantified.
- **Set A was changing while it was measured.** Its bands drifted 8–37 % over 32 minutes. The fit treats its six runs as six samples of one state; they are not.

13 · What would settle it

Four tests. **T0 has been done — chapter 9. The rest have not.**

T0 — carry r_Q onto a pair it was not fitted on. ✓ DONE (chapter 9), and it passed once and failed twice. Within a rig state the correction did exactly what it promised; across a rebuild it did the opposite. That result is what makes T1 worth doing and T1b necessary.

T1 — measure r_Q on every oil in the campaign. Each oil needs at least two preparations of genuinely different strength, ideally spanning more than Steirerkraft's narrow 17 %. Then ask the only question that remains: **within one rig state, is r_Q one constant, or does each sample have its own?** If the four values agree within their errors, A1 is supported well enough to adopt. If they scatter, the correction is dead in this form. **This costs nothing beyond what the campaign is already scheduled to do**, provided each oil is prepared at two strengths rather than one.

T1b — re-measure r_Q after the next mechanical change, and check it moved. Chapter 9 says it should. If it does not move when the optics change, the whole physical story is wrong and the value is fitting something else. This is a cheap and unusually sharp test, because it predicts a *change* rather than a constancy — and a prediction of change is much harder to satisfy by accident.

T2 — report both numbers in parallel. Compute M and M_∞ on every run and record both. The corrected number changes no decision while it is only being recorded, and after a dozen samples its behaviour is known rather than argued about.

T3 — attack r_Q physically instead. If the sample can be made genuinely clear — 0.22 μm filtration, or a solvent that dissolves the oil rather than suspending it — then r_Q goes toward zero, and no arithmetic correction is needed at all. **This is the better outcome**, and it is the strongest remaining argument for the filtration work: not that it improves precision, but that it removes the term that makes the number non-transferable.

The recommendation of this document is T2 immediately, T1 during the campaign, T1b at the next rig change — and explicitly not to change the shipped verdict.

Appendix A — every number

Produced by `diagnostics/pedestal_correction.py` . Set values are means over runs.

set	n	A_Soret raw	A_Q raw	turbidity 520–540	B_Soret	B_Q	M shipped
Kiendler A	6	0.8263	0.1108	0.0378	0.8221	0.0576	14.279
Kiendler B	2	1.1231	0.1994	0.0919	1.0819	0.0849	12.740
Kiendler C	2	1.1705	0.2082	0.1018	1.1160	0.0856	13.039
Steirerkraft B	6	1.0931	0.1968	0.0981	1.0331	0.0828	12.489
Steirerkraft C	6	1.1864	0.2300	0.1231	1.1043	0.0901	12.251
S-Budget D	6	1.0855	0.2251	0.1038	1.0093	0.1085	9.303

Provenance. Kiendler = `spectracs-references/tmp/20260801A|B|C/` , one evening, three preparations of one oil. Steirerkraft = `20270729B/C` , two fills. S-Budget = `20260731A` , six re-seats of one fill ("series D"). All post-rebuild, same instrument and protocol.

Constants used. `r_Q` = -0.0246 A (Kiendler, run-level fit). Bands: Soret 440–460, Q 560–580, anchors 520–540 and 600–630 nm.

Chapter 9's dilution pairs. `oilK / oilL` = one green oil at 2 and 3 drops; `oilN / oilM` = one brown oil at 2 and 3 drops (both 2026-07-21, PRE-rebuild); `20270729B / C` = Steirerkraft at two strengths, post-rebuild. Produced by `diagnostics/all_metrics_archive.py` , which also emits the whole archive — 122 measurements across 40 series — as a single CSV.

Appendix B — reproduction

```
export PYTHONPATH="./diagnostics:../spectracsPy-core:../spectracsPy-model:../spectracsPy-base:../spectracsPy-server:../spectracs-plugins"

./venv/bin/python diagnostics/pedestal_correction.py      # chapters 1-7: numbers + both figures
./venv/bin/python diagnostics/all_metrics_archive.py      # chapter 9: the out-of-sample test

python3 docs/tools/build_pedestal_correction_pdf.py      # rebuild this PDF
```

The first prints every table in chapters 1–7 and writes both figures into `docs/figures/` . The second prints chapter 9's dilution-pair table, plus the whole-archive metric matrix it is drawn from. The third rebuilds this PDF from the markdown master.

Appendix C — where this sits in the specifications

the derivation this confirms	<code>SPEC_capture_quality.md</code> §16.14.4–6 — pedestal curvature as the <i>only</i> mechanism that can break dilution invariance
the session that measured it	§16.15, and the lab diary entry for 2026-08-01/02

the bound this supersedes	§16.14.7 asserted a bound of 0.008 A on the residual from pooled archive data; the measured value is $\sim 3\times$ that
the threshold at risk	§16.10.17d, and §16.15.7 for what the pedestal-free scale does to it
the open question	§16.10.8, dilution invariance, which the 2026-08-01 session did not close